

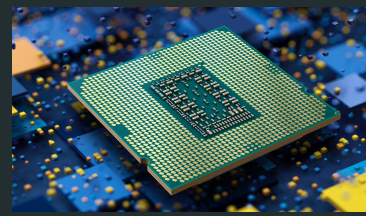
Project Proposal for RISC-V Vector Extension

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What is A CPU?



- Central Processing Unit (CPU): The main chip in computers that handles everything from user input to math computations. Basically the brain of the computer. Each chip uses an ISA.
- Instruction Set Architecture (ISA): The abstract model that is the interface between hardware and software. Runs assembly code. Examples include x86 (used by Intel and AMD) and ARM (used in most mobile devices and by Apple, Qualcomm, Mediatek etc.)
 - Allows software and hardware developers to target a single standard instead of everyone doing their own thing
- Assembly: Also known as machine code. It is the code that coding languages like C, Rust, and Go are turned into so that the machine can understand and execute the code.

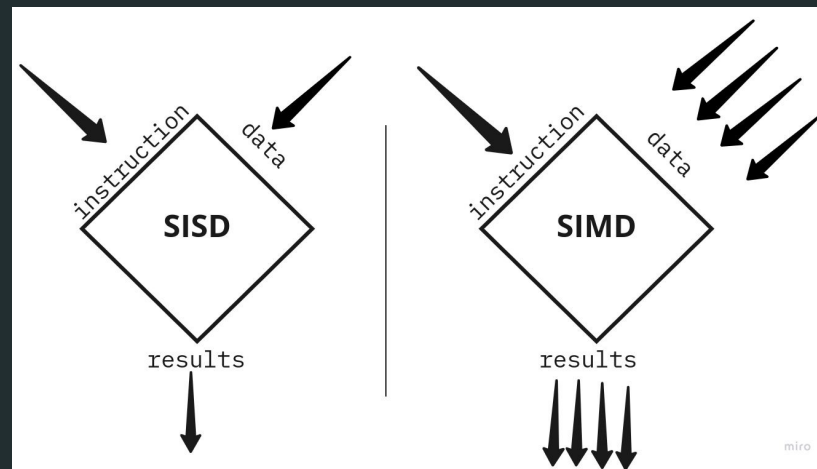
What is RISC-V?

- Reduced Instruction Set Computing, 5th Generation (RISC-V)
- Open source CPU Architecture
- Developed in UC Berkeley in 2010, ratified in 2016
 - With an extended set stabilized last year
- No royalties or legal issues
- Gaining traction, especially by hackers, students, academics, due to accessibility
 - As well as wider industry
- Reduced Instruction Set makes it easier to implement and more flexible



What is a Vector Extension?

- Extensions to a cpu that allow doing the same operation on multiple pieces of data in parallel
- More data = higher throughput = faster speed
- Also known as SIMD (Single Instruction Multiple Data)
- RISC-V vector processing is unique in that it is not a fixed size and can change at runtime
 - Newer hardware can automatically scale better
- 1.0 spec is currently being ratified



Problem Statement

Create an Open Source RISC-V Vector extension written in Verilog that supports floating point and complex number math.

Required Tools:



Verilog: An industry standard Hardware Description Language (HDL): A language for designing chips and other hardware.

Field Programmable Gate Array (FPGA): Reconfigurable/programmable chips unlike the chips in your phone. This will allow us to see if our project can be made into an actual fabricated chip.

Vivado: The tool required to take our design and put it onto an FPGA (Specifically Xilinx FPGAs)

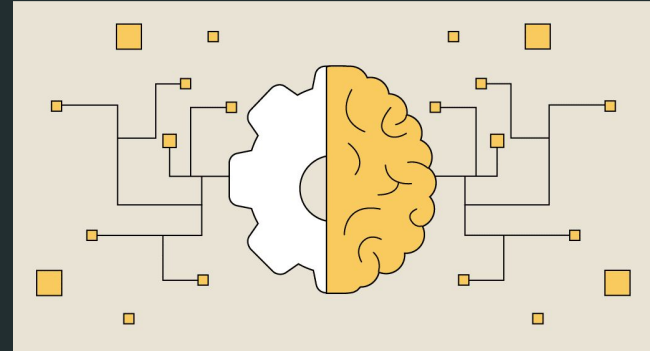
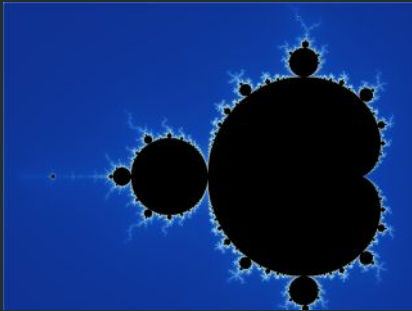
Verilator/Synopsys/IVerilog/Yosys : Tools required to simulate our hardware design.

GCC RISC-V compiler: A tool used to take C code and translate it into assembly

Open Source CPU design: Since designing a CPU is its own (much larger) project and there simply isn't enough time to build one from scratch.

Uses of a Vector Extension co-processor

- Useful for workloads that require processing large throughputs of data
- Used in computer simulations to run mathematical models such as fractals (Mandelbrot Set)
- Used in training Machine Learning (ML) models on a CPU which requires many matrix-vector calculations



Proposed Method

1. Vet open source RISC-V CPUs to use as a base for our project
 - a. Needs to fit on our FPGA and have the bare minimum features to limit unknown variables.
2. Study the RISC-V Vector Extension spec and try to implement it
 - a. Implement integer math before floating point and complex arithmetic.
 - i. Floating points = decimal number (with limited precision/error)
 - b. We will avoid imprecise traps for simplicity and force precise traps
 - i. Precise traps: Allows errors partially through an operation
 - ii. Imprecise trap: operations must complete and be in a valid state
3. We will then get the CPU on the FPGA and try to run code on it

Current Progress

SECTION NUMBER	TASK TITLE	TASK OWNER	START DATE	DUE DATE	DURATION	PCT OF TASK COMPLETE
1	Minimum Requirements					
1.1	Get CPU on FPGAs	Khondakar	9/23/24	10/6/24	13	100%
1.2.1	Evaluate CPU choices	Muhammad	9/30/24	11/17/24	47	100%
1.2.2	Run Synthesized CPU on FPGA	Khondakar	10/7/24	12/6/24	59	50%
1.2.3	Pass all testbenches	Khondakar, Niels, Muhammad	10/14/24	12/6/24	52	75%
1.3.1	Get familiar with Load/Store	Niels, Khondakar	10/13/24	12/6/24	53	75%
1.3.2	Run Mandelbrot Set on chosen CPU	Muhammad	11/17/24	12/6/24	19	50%
1.3.3	Implement Vector Arithmetic Instructions	Niels	11/4/24	12/6/24	32	40%
1.4.1	Implement Vector load/store				0	0%
1.4.2	Implement exception handling/hazard detection				0	0%
1.4.3	Implement Vector Mask Instructions				0	0%
1.5	Implement Vector Fixed-Point arithmetic Instructions				0	0%
1.6.1	Implement Vector Configuration Setting Instructions				0	0%

Current Progress (Gantt Chart)

[illegible]

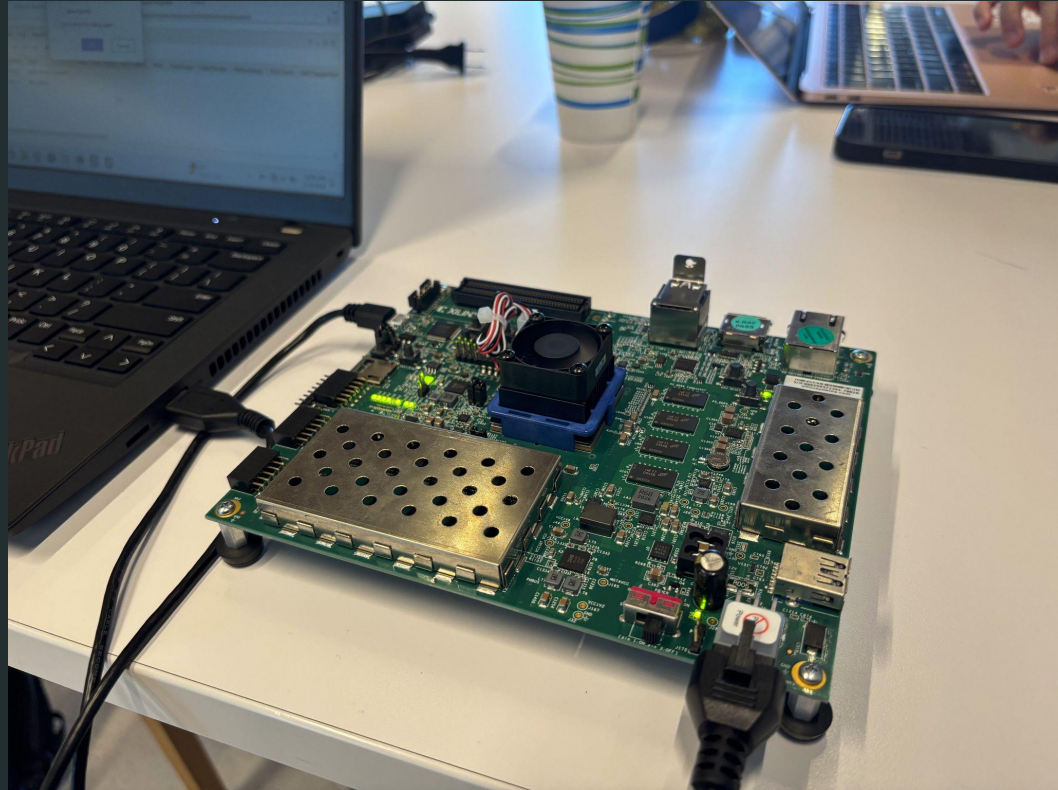
Current Progress Continued

- We initially settled on using a RISC-V core from UltraEmbedded to build on
 - Very Modular
 - Easy to understand Design
 - Fits on our FPGA (ZCU104 Xilinx Evaluation Board)
- The simulation environment for this core requires a deprecated version of Verilator which is not available (and we had difficulty compiling)
- We then decided to vet other cores and now decided to use the PicoRV32 RISC-V core
 - It fits on our FPGA
 - We were able to setup the simulation environment and was able to run code on it in simulation
- Now we are currently working on sample programs and getting it on the FPGA

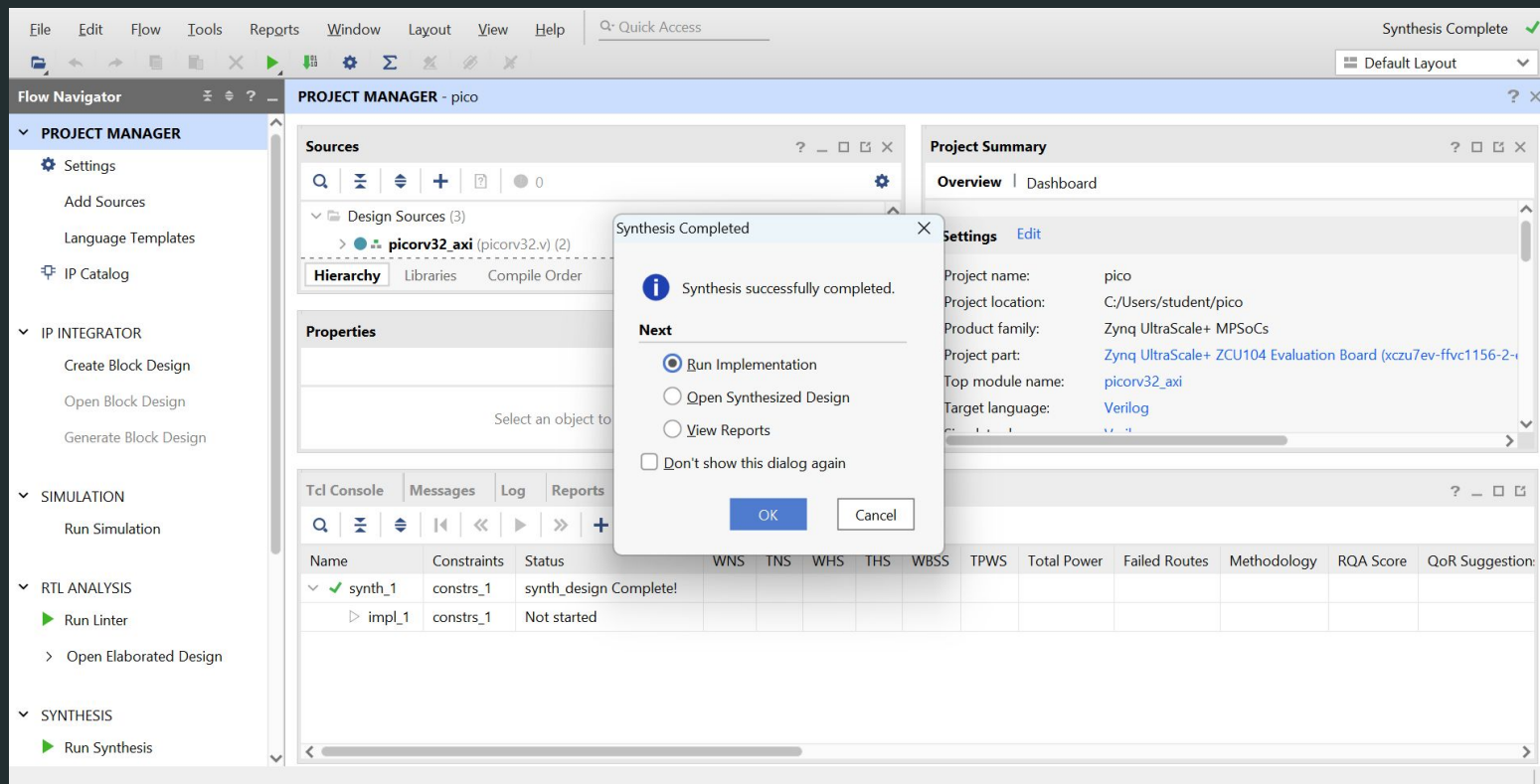
Current Progress

[illegible]

```
TRAP
- testbench.v:112: Verilog $finish
```



Current Progress



Current Progress

The screenshot displays the Xilinx Vivado IDE interface. A central dialog box titled "Implementation Completed" is open, indicating that the implementation process has finished successfully. The dialog provides three main options for the user to proceed:

- ☒ Open Implemented Design
- ☐ Generate Bitstream
- ☐ View Reports

There is also a checkbox for "Don't show this dialog again" which is currently unchecked. The "OK" button is highlighted in blue.

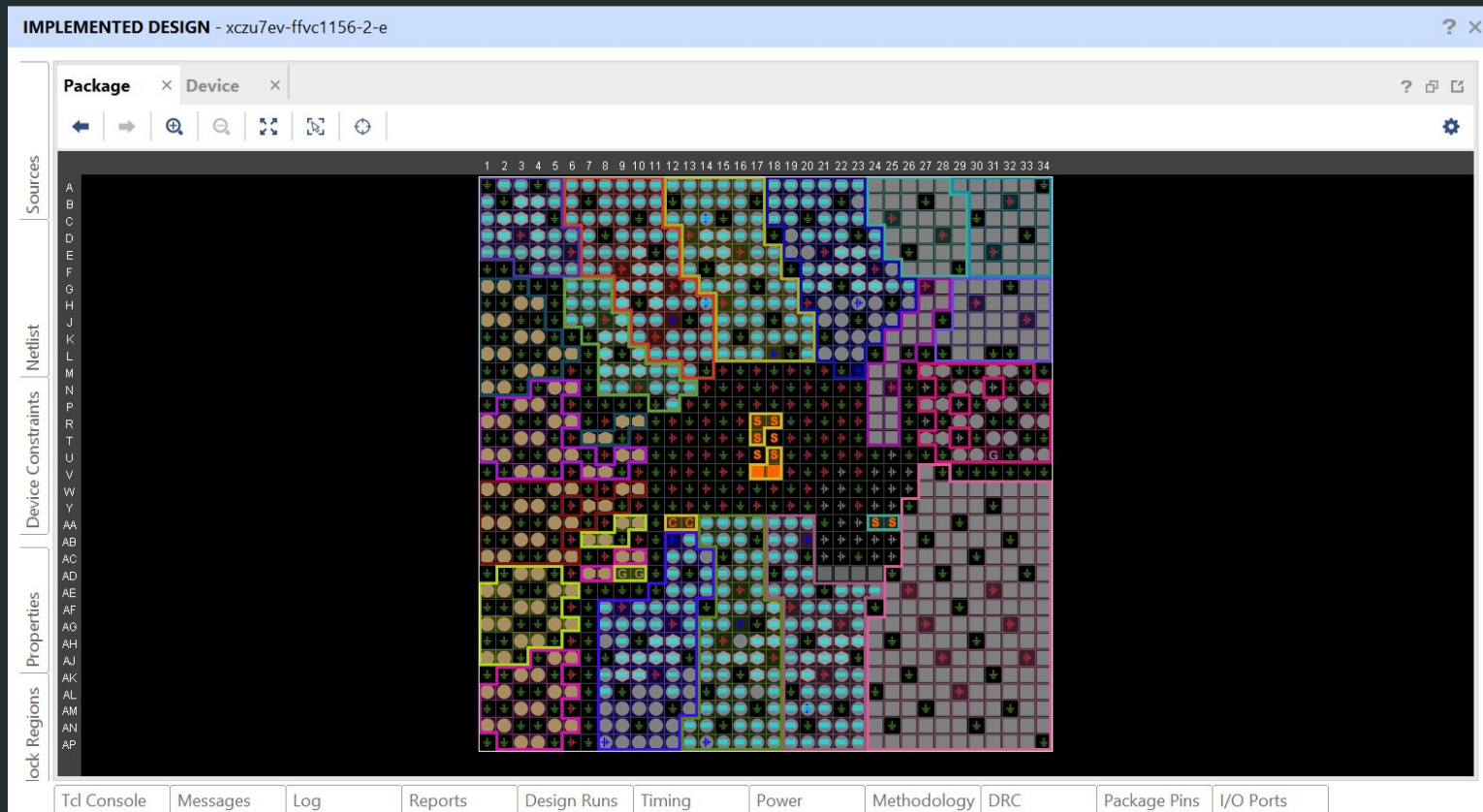
The background interface shows the "PROJECT MANAGER - pico" window. The "Sources" pane lists "Design Sources (3)" including "picorv32_axi (picorv32.v) (2)". The "Properties" pane is empty. The "Tcl Console" pane shows the following table:

Name	Constraints	Status	WNS	TNS	WHS	THS	WBSS	TPWS	Total Power	Failed Routes	Methodology	RQA Score	QoR Sugg
✓ synth_1	constrs_1	synth_design Complete!											
✓ impl_1	constrs_1	route_design Complete!	NA	NA	NA	NA		NA	38.066	0	726 CW		

The "Project Summary" window is also visible, showing the "Overview" tab with the following details:

- Project name: pico
- Project location: C:/Users/student/pico
- Product family: Zynq UltraScale+ MPSoCs
- Project part: Zynq UltraScale+ ZCU104 Evaluation Board (xczu7ev-ffvc1156-2-)
- Module name: picorv32_axi
- Target language: Verilog

Current Progress



Current Progress

IMPLEMENTED DESIGN - xczu7ev-ffvc1156-2-e

Package x Device x

Navigation icons: back, forward, search, zoom, pan, rotate, etc.

Sources

Netlist

Device Constraints

Properties

Lock Regions

Td Console Messages Log Reports Design Runs Timing Power Methodology DRC Package Pins I/O Ports

MVP vs Target Vs Stretch Goals

MVP:

Add a Vector Extension Co-Processor to a RISC-V CPU that can do integer math

Target:

Add support for floating point complex number math

Stretch goal:

Support for imprecise traps

More speed optimizations

Hazards?

No Hazards

- Everything is on a computer
- Mild electrical issues

Anticipated Impact

1. A freely available and open source implementation that allows others to study, modify and use it
2. Reduced runtimes of compiled code utilizing vector-based operations
3. Companies that require high throughput and parallel processing like machine learning companies aren't stuck with a single vendor like NVIDIA and AMD who can change their pricing on a whim.

Bibliography & References

- <https://github.com/riscvarchive/riscv-v-spec/releases/tag/v1.0>
- <https://github.com/YosysHQ/picorv32>
- <https://github.com/ultraembedded/riscv>

Extra Slide 1

197 Presentation Feedback:

- explaining the information more clearly to non-electrical engineering students
- longer background
- explain what the vector extension is used for
- add images
- imprecise traps (better explain that)
- anticipated outcome: what impact does it have?
- clarify the requirements and priorities

Extra Slide 2

Based on the feedback we received from our draft presentation, two of the big takeaways we got was revising the presentation so a general audience (people with little to no computer knowledge) can follow along and identifying an issue our project is trying to solve. Taking these into consideration, we'll revise our final presentation with these changes to leave a lasting impression on the audience.